POWER PULSE

Power Flow Through a Transmission Line Important Formulae

Notes by Balaji Sir

1 General (ABCD) form

1. $$S_r = \frac{|V_s||V_r|}{|B|}\angle(\beta-\delta) - \frac{|A||V_r|^2}{|B|}\angle(\beta-\alpha)$$

A = |A|∠α, B = |B|∠β, δ = angle of V_s over V_r

2. $$P_r = \frac{|V_sV_r|}{|B|}\cos(\beta-\delta) - \frac{|A||V_r|^2}{|B|}\cos(\beta-\alpha)$$

3. $$Q_r = \frac{|V_sV_r|}{|B|}\sin(\beta-\delta) - \frac{|A||V_r|^2}{|B|}\sin(\beta-\alpha)$$

4. $$P_{r,max} \text{ at } \delta=\beta: \ \frac{|V_sV_r|}{|B|} - \frac{|A||V_r|^2}{|B|}\cos(\beta-\alpha)$$

2 Lossless line (R = 0)

5. $$P = \frac{|V_s||V_r|}{X}\sin\delta, \qquad P_{max} = \frac{|V_s||V_r|}{X} \ (\delta=90^\circ)$$

three-phase: use line-to-line voltages

6. $$Q_r = \frac{|V_r|}{X}\big(|V_s|\cos\delta - |V_r|\big), \quad Q_s = \frac{|V_s|}{X}\big(|V_s| - |V_r|\cos\delta\big)$$

7. $$Q \approx \frac{|V|\,\Delta|V|}{X} \ (\delta \text{ small})$$

P is controlled mainly by δ, Q mainly by voltage difference

8. $$X_{eff} = X(1-k_{se}), \qquad P_{max} \propto \frac{1}{X_{eff}}$$

series compensation

9. $$P_{loss} = P_s - P_r = 3I^2R$$

10. $$\frac{dP}{d\delta} = P_{max}\cos\delta > 0$$

stable operation (steady-state stability limit at δ = 90°)